

PAVAN REDDY PULUSANI

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PROFESSIONAL SUMMARY

Results-driven Data Science professional with 2+ years of combined experience across cloud operations, data analytics, and machine learning. Proven ability to design scalable data pipelines, develop predictive models, and deploy AI-driven solutions using Python, SQL, and cloud platforms such as AWS. Skilled in leveraging tools like Pandas, Scikit-learn, TensorFlow, and Power BI to extract insights and drive business outcomes. Adept at implementing NLP, computer vision, and statistical modeling techniques to solve real-world problems. Strong foundation in MLOps, data engineering, and dashboard development, with a track record of delivering end-to-end solutions from data ingestion to model deployment and visualization.

TECHNICAL SKILLS

Programming Languages: Python (NumPy, Pandas, Scikit-learn, Py Torch, TensorFlow, XG Boost, Light (GBM), R, SQL, Bash, Java, C++

Data Manipulation & Analysis: Pandas, NumPy, Dask, Polars, SQL (PostgreSQL, MySQL), Apache Spark (PySpark), Google Big Query

Machine Learning & Deep Learning: Scikit-learn, TensorFlow, Keras, Py Torch, Hugging Face Transformers, Auto ML, ML flow, Optuna, Light GBM, XG Boost, Cat Boost

NLP & Computer Vision: SpaCy, NLTK, OpenCV, YOLOv8, Transformers (BERT, GPT), Lang Chain

Data Visualization: Power BI, Tableau, Plotly, Seaborn, Matplotlib, Dash, Streamlit

MLOps & Model Deployment: Docker, Kubernetes, ML flow, Airflow, DVC, Fast API, Flask, ONNX, Torch Script

Cloud Platforms & Tools: AWS (S3, EC2, SageMaker, Lambda, Glue, Athena), GCP (Vertex AI, Big Query, Cloud Functions), Azure (ML Studio, Blob Storage, Synapse, Azure ML)

Version Control & DevOps: Git, GitHub Actions, Jenkins, Terraform, CI/CD pipelines

Databases: PostgreSQL, MySQL, MongoDB, Cassandra, Redis

Big Data & Pipelines: Apache Kafka, Apache Airflow, Hadoop, Spark, Databricks, Data Bricks Auto ML

Other Tools: Jupyter Lab, VS Code, Google Colab, Notebooks, Weights & Biases, Stream lit, Excel (PivotTables, Macros), REST APIs

WORK EXPERIENCE

Data Science Intern

Sep 2024 - Feb 2025

Clover soft LLC ,Raleigh, North Carolina

- Encountered challenges with unstructured client feedback data; developed NLP pipelines using Python, SpaCy, and NLTK to extract sentiment and topics, enabling real-time insight generation and stakeholder reporting.
- Struggled with inconsistent data pipelines across departments; implemented modular ETL workflows using Pandas, SQL, and Airflow to automate data ingestion and standardize transformation logic across sources.
- Faced inefficiencies in model performance tuning; applied hyperparameter optimization using GridSearchCV and Randomized SearchCV in Scikit-learn to improve classification model accuracy by 18%.
- Identified bottlenecks in exploratory data analysis; leveraged Seaborn and Plotly to visualize high-dimensional data patterns and anomalies, enhancing business decision support.
- Required real-time dashboards for executive reporting; built Power BI dashboards integrated with SQL Server to deliver interactive KPIs and drill-down insights on customer churn and product usage.
- Addressed manual effort in feature engineering; created reusable Python scripts to automate variable selection, encoding, and scaling, accelerating model deployment cycles.
- Encountered issues in productionizing ML models; containerized solutions using Docker and deployed RESTful APIs via Flask to enable scalable model consumption by external applications.
- Noticed delays in cross-team collaboration; set up Jupyter Notebook documentation with Markdown and Git for reproducible experiments and seamless model handoff between data science and engineering teams.

Technical Support Engineer - Data & Cloud Operations

Mar 2022 - Jul 2023

Grey Campus, Hyderabad, India

- Diagnosed recurring system latency issues by analyzing AWS CloudWatch logs and EC2 performance metrics, then configured Auto Scaling Groups and load balancers to improve system uptime to 99.9%.
- Faced delays in resolving repetitive operational issues, so developed Python automation scripts using Boto3 to monitor resource health and trigger alerts, reducing manual effort by over 40%.
- Encountered inefficiencies in log processing workflows; deployed AWS Lambda with S3 triggers to automate log ingestion and real-time error flagging, accelerating anomaly detection and root cause analysis.
- Dealt with inconsistencies in infrastructure setup across environments; implemented AWS CloudFormation templates and Git-based version control to enable repeatable, error-free deployments.
- Observed rising cloud spend due to underutilized compute resources; conducted AWS Cost Explorer analysis and implemented EC2 right-sizing and RDS storage optimization, cutting costs by 20%.
- Identified security gaps in S3 bucket access and IAM configurations; enforced least-privilege IAM policies, enabled S3 encryption, and applied lifecycle rules to secure sensitive data and reduce storage overhead.
- Lacked visibility into system usage patterns for leadership reporting; connected AWS logs to Power BI and designed dashboards showing compute usage, user activity, and incident trends for strategic decisions.
- Faced high ticket volume due to repetitive support queries; created automated knowledge base workflows and Python scripts for first-level triaging, improving support resolution time and user satisfaction.

Data Science Intern

Apr 2020 - Sep 2020

CureFit (Cult Fit), Hyderabad, India

- Observed inconsistency in user workout adherence; built time-series models using Prophet and ARIMA in Python to forecast user drop-offs and suggest personalized session timing, improving retention rates by 22%.
- Faced a challenge in targeting promotions; performed customer segmentation using K-Means clustering on user behavior and purchase history, enabling data-driven marketing campaigns that increased conversion rates by 15%.
- Encountered missing values and irregularities in fitness tracking data; implemented robust data cleaning workflows using Pandas and SQL, enhancing data quality and model reliability.
- Needed real-time performance monitoring; developed dashboards in Tableau and Power BI to visualize workout completion rates, calorie burn estimates, and churn metrics across fitness centers.
- Required scalable ML pipelines; created modular scripts in Python with Scikit-learn for automating model training, validation, and deployment, cutting model refresh time by 40%.
- Identified inefficiencies in content recommendation; engineered features based on session feedback and built collaborative filtering models to improve workout and nutrition plan suggestions.
- Found performance drop in models with new data; designed continuous learning architecture using scheduled retraining with updated data pipelines and versioned datasets in Git.
- Collaborated with cross-functional teams including mobile devs and product analysts to deploy predictive insights into the Cult Fit mobile app, driving personalized fitness engagement for over 10,000 users.

PROJECTS

Healthcare Analytics Decision Support System (Python, SQL, Power BI, Pandas, Seaborn)

- Faced challenges with fragmented community health data; developed ETL workflows using Python (Pandas) and SQL to clean, standardize, and validate datasets before loading into Power BI for centralized analysis.
- Identified inefficiencies in healthcare delivery; built classification and regression models to forecast high-risk populations and predict future care demand, aiding proactive planning.
- Designed interactive dashboards in Power BI and Seaborn to present program KPIs and patient outcomes to health directors, supporting data-driven resource allocation.

Facial Recognition-Based Attendance System (Python, OpenCV, Excel Automation)

- Manual attendance tracking lacked efficiency and accuracy; engineered a facial recognition system using OpenCV and Haar cascades to automate real-time attendance logging via live camera feeds.
- Built a custom dataset and trained recognition models for unique face detection; implemented Excel integration using Python to log timestamps, class, and student identity.
- Enabled administrative reporting and audit trails by generating automated attendance sheets, significantly reducing human error and administrative workload.

AWS Static Website Deployment & Monitoring (AWS S3, CloudFront, Route 53, IAM, CloudWatch)

- Required a scalable and cost-effective hosting solution; deployed a static website on AWS S3 with CloudFront for global content delivery, improving page load performance by 40%.
- Configured IAM policies and Route 53 for secure domain management and DNS routing; enforced bucket policies for restricted access and SSL encryption.
- Set up real-time monitoring using AWS CloudWatch and created automated alerts to track traffic spikes and server errors, enhancing reliability and observability.

Ticketing System Optimization for Internal Support (Jira, AWS Console, Python)

- Identified bottlenecks in support workflows; developed Python scripts to automate ticket categorization, assignment, and status updates via internal APIs.
- Integrated error log scanning to detect recurring issues and applied structured escalation rules, reducing resolution time by 35%.
- Visualized resolution trends and backlog metrics using Python charts to improve transparency and drive data-informed support decisions.

EDUCATION

Master of Science in Computer Science

Saint Louis University, St. Louis, MO

Bachelor of Applied Science in Electronics and Communication Engineering

MVSR Engineering College, Hyderabad, India

CERTIFICATIONS

- AWS Academy Cloud Foundations (Verified)
- CCNA: Introduction to Networks – Virtual Internship
- Hands-on Python Training – STEM Innovation Technologies